ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

MID SEMESTER EXAMINATION **SUMMER SEMESTER, 2024-2025**

DURATION: 1 HOUR 30 MINUTES **FULL MARKS: 75**

Math 4641: Numerical Methods

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 3 (three)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

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1. a) How can you broadly categorize root-finding algorithms? Compare and contrast the advantages and disadvantages of these categories. $1 + 3$ (CO4) (PO1)

Solution:

Any 2 valid points of comparison will suffice. Some comparisons can be written as follows —

Criteria	Bracketing Methods	Open Methods
Initial guesses	Need two guesses x_l, x_u that bracket the root (i.e. $f(x_l)f(x_u) < 0$).	Need one (or two) starting guesses that need not bracket the root.
Convergence	Always converge (guaranteed), but slowly (linearly).	Faster (superlinear/quadratic), but may diverge.
Examples	Bisection, False Position.	Newton–Raphson, Secant.

Rubric:

- 1 point for the categorization.
- 1.5 points for each valid point of comparison.

b) Derive the formula for the Newton–Raphson method geometrically. 6 (CO4) (PO1)

Solution:

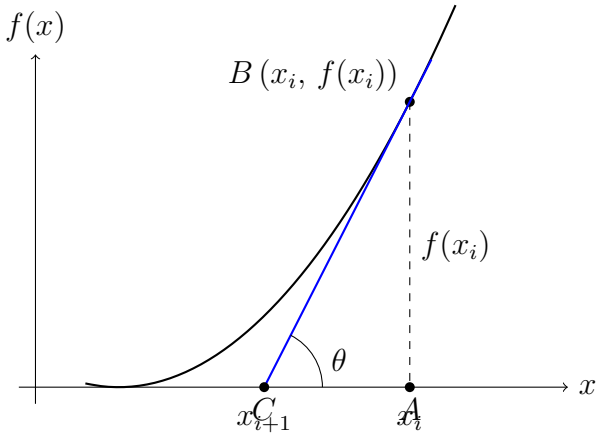


Figure 1: Geometrical representation of the Newton–Raphson method.

The tangent to $f(x)$ drawn at the point $B\,(x_i, f(x_i))$ has a slope equal to $f'(x_i)$, and it intersects the x-axis at $C\,(x_{i+1}, 0)$.

From the right-angled triangle $\triangle BAC$,

$$\tan(\theta) = \frac{AB}{AC}$$
$$f'(x_i) = \frac{f(x_i)}{x_i - x_{i+1}}$$
$$\Rightarrow x_i - x_{i+1} = \frac{f(x_i)}{f'(x_i)}$$
$$\Rightarrow x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

Hence, we get the formula for the Newton–Raphson method,

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

Rubric:

- 4 points for a correct geometric derivation.
- 2 points for the correct formula.

c) Suppose, your team is awarded a direct free-kick $x = 25$ m straight out from the centre of the goal. The defending wall blocks the direct shot, so the striker kicks the ball out to the right at an initial lateral angle θ and applies heavy counter-clockwise spin. The resulting Magnus force curves the ball back to the left in mid-air, and the striker wants it to strike the inside of the left goalpost, at a lateral position $z = -3.6$ m from the centre line.

Accounting for the linear air drag that slows the ball's forward velocity, the lateral position z of the ball as a function of the forward distance x is modelled by the transcendental equation

$$z = x\tan(\theta) + \frac{S}{c^2}\left[cx + v_0\cos(\theta)\ln\left(1 - \frac{cx}{v_0\cos(\theta)}\right)\right] \tag{1}$$

where $v_0 = 30$ m/s (initial speed), $c = 0.15\,\text{s}^{-1}$ (linear drag coefficient per unit mass), $S = 0.65\,\text{m/s}^2$ (the Magnus spin parameter), and θ is the unknown initial launch angle in radians.

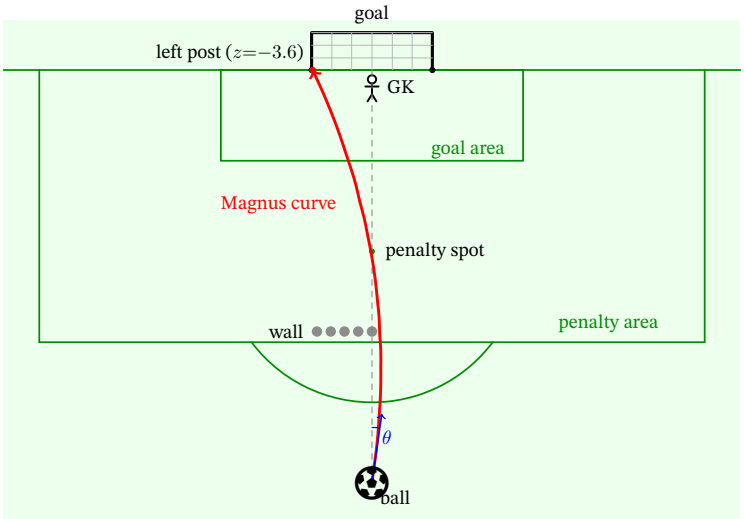


Figure 2: Bird's-eye view of the curved free-kick for Question 1.c.

i. Formulate a function $f(\theta)$ such that its root will yield the lateral launch angle θ at which the ball strikes the inside of the left goalpost. 2 (CO1) (PO2)

Solution:

The ball strikes the right post when $z = -3.6\,\text{m}$ at the forward distance $x = 25\,\text{m}$. Moving everything to one side of the modelling equation,

$$f(\theta) = x\tan(\theta) + \frac{S}{c^2}\left[cx + v_0\cos(\theta)\ln\left(1 - \frac{cx}{v_0\cos(\theta)}\right)\right] - z_{\text{target}}$$

Plugging in $x = 25$, $z_{\text{target}} = -3.6$, $v_0 = 30$, $c = 0.15$ (so $cx = 3.75$ and $\frac{S}{c^2} = 28.889$),

$$f(\theta) = 25\tan(\theta) + 28.889\left[3.75 + 30\cos(\theta)\ln\left(1 - \frac{3.75}{30\cos(\theta)}\right)\right] + 3.6$$

Rubric:

- 2 points for the correct function $f(\theta)$.

ii. Apply the Bisection method to estimate the lateral launch angle θ (in radians). The initial guesses are $\theta_l = 0.12$ and $\theta_u = 0.18$. Demonstrating the step-by-step mathematical procedure, conduct 4 iterations, and find the relative approximate error ($|\epsilon_a|\%$) and the number of significant digits that are at least correct (m) at the end of each iteration. Draw Table 1 and fill it out after performing the necessary calculations. 10 (CO2) (PO1)

Table 1: The relevant values obtained in the answer of Question 1.c

Iteration	θ_l	θ_u	θ_m	$\lvert\epsilon_a\rvert\%$	m	$f(\theta_m)$
1	0.12	0.18		–	–	
2						
3						
4						

Solution:

At each step the midpoint is $\theta_m = \dfrac{\theta_l + \theta_u}{2}$. If $f(\theta_l)\cdot f(\theta_m) < 0$ the root lies in $[\theta_l, \theta_m]$, so we set $\theta_u = \theta_m$; otherwise we set $\theta_l = \theta_m$. Using $f(\theta) = 25\tan(\theta) + 28.889\left[3.75 + 30\cos(\theta)\ln\left(1 - \frac{3.75}{30\cos(\theta)}\right)\right] + 3.6$, this yields the following table.

Iteration	θ_l	θ_u	θ_m	$\lvert\epsilon_a\rvert\%$	m	$f(\theta_m)$
1	0.12	0.18	0.15	–	–	−0.107378
2	0.15	0.18	0.165	9.0909%	0	0.257546
3	0.15	0.165	0.1575	4.7619%	1	0.075099
4	0.15	0.1575	0.15375	2.4390%	1	−0.016135

After 4 iterations the root is bracketed in $[0.15375, 0.1575]$ (true value $\theta \approx 0.154413$ rad $\approx 8.85°$), and the estimate is correct to at least 1 significant digit ($m = 1$). The striker should aim about $8.85°$ to the left and let the Magnus curl bend the ball into the right post.

Note: The values above are highly precise and are not indicative of other validly approximated answers.

Rubric:

- 0.5 point for each approximately correct answer (rounded down to the nearest 0.5).
- 1 grace point for demonstrating the step-by-step procedure in each iteration.

2. a) Define *Round-off Error* and *Truncation Error*. With one proper example for each, explain how it is introduced. $2+2$ (CO4) (PO1)

Solution:

Round-off Error is the error that arises because a computer can only store numbers with a finite number of significant digits, so the remaining digits are dropped (rounded/-chopped).
Example: Storing the value of $\frac{1}{3}$ as 0.3333 discards the infinitely repeating tail, introducing an error of $\frac{1}{3} - 0.3333 = 3.33\ldots \times 10^{-5}$.

Truncation Error is the error caused by approximating an exact mathematical process by a finite/approximate one.
Example: Keeping only the first 3 terms of the Maclaurin series of e^x gives $e^x \approx 1 + x + \frac{x^2}{2!}$, and the truncation error is $e^x - \left(1 + x + \frac{x^2}{2!}\right)$.

Rubric:

- 1 point for each correct definition.
- 1 point for each valid example.

b) The Maclaurin series for $\sin(x)$ is given as follows:

$$\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$$

i. Why is the sinusoid function $\sin(x)$ categorized as a transcendental function? 1 (CO4) (PO1)

Solution:

The sinusoid function $\sin(x)$ cannot be expressed using a finite sequence of terms or algebraic operations, which is why it falls under the umbrella of transcendental functions.

Rubric:

- 1 point for the correct reason.

ii. Use the Remainder Theorem to find bounds on the truncation error in the representation of $\sin(2)$ if only the first 3 terms of the Maclaurin series are used. 5 (CO2) (PO1)

Solution:

We approximate $\sin(2)$ by the first three nonzero terms of its Maclaurin series

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots.$$

We know,

$$\begin{aligned} f(0) &= \sin(0) = 0 \\ f'(0) &= \cos(0) = 1 \\ f''(0) &= -\sin(0) = 0 \\ f^{(3)}(0) &= -\cos(0) = -1 \\ f^{(4)}(0) &= \sin(0) = 0 \\ f^{(5)}(0) &= \cos(0) = 1 \end{aligned}$$

So, the first 6 terms of the Taylor series needs to be used to get the first 3 terms of the Maclaurin series, in the case of $\sin(x)$. The truncation (remainder) after these three terms at $x = 2$ is the remainder $R_5(2)$.
By the Lagrange form of the remainder,

$$R_5(0+2) = \frac{f^{(5+1)}(c)}{(5+1)!}\,2^{(5+1)}$$

for some c between 0 and 2. For $f(x) = \sin x$ the derivatives cycle; in particular

$$f^{(6)}(x) = -\sin x.$$

Hence

$$R_5(2) = \frac{-\sin(c)\,2^6}{6!} = -\frac{64\sin(c)}{720} = -\frac{4}{45}\sin(c), \qquad \text{for some } 0 < c < 2.$$

Since $0 < c < 2$ and $|\sin(c)| \le 1$, we obtain the magnitude bound

$$\left|R_5(2)\right| \le \frac{4}{45} \approx 0.088888\ldots.$$

Moreover, because $0 < c < \pi$ (and $2 < \pi$), we have $\sin(c) > 0$, so $R_5(2)$ is negative and

$$-\frac{4}{45} < R_5(2) < 0.$$

Rubric:

- 2 points for the correct Remainder Theorem Lagrange form.
- 3 points for the correct bounds.

c) The Taylor series for a function $f(x)$ is — 15 (CO4) (PO1)

$$f(x+h) = f(x) + f'(x)\frac{h}{1!} + f''(x)\frac{h^2}{2!} + f'''(x)\frac{h^3}{3!} + f''''(x)\frac{h^4}{4!} + f'''''(x)\frac{h^5}{5!} + \ldots \quad (2)$$

Using the Taylor series (with at least the first 5 terms), derive the Maclaurin series of

$$f(x) = x \cdot \operatorname{cosec}(x)$$

Solution:

The Maclaurin series of a transcendental function is simply a Taylor series around the point $x = 0$. For this problem, let's handle the $\operatorname{cosec}(x)$ term first.
We know, $\frac{d}{dx}(\operatorname{cosec}(x)) = -\operatorname{cosec}(x)\cot(x)$, and $\operatorname{cosec}(0) =$ undefined. So we will get undefined values for $f(0)$ and $f^i(0)$.
Instead let's consider, $f(x) = \operatorname{cosec}(x) = \dfrac{1}{\sin(x)}$, and find the Maclaurin series of the denominator $g(x) = \sin(x)$. Let's calculate its derivatives at the point $x = 0$.

$$\begin{aligned}
g(x) &= \sin(x), \quad & g(0) &= 0\\
g'(x) &= \cos(x), \quad & g'(0) &= 1\\
g''(x) &= -\sin(x), \quad & g''(0) &= 0\\
g'''(x) &= -\cos(x), \quad & g'''(0) &= -1\\
g''''(x) &= \sin(x), \quad & g''''(0) &= 0
\end{aligned}$$

Using the first 5 terms of the Taylor series,

$$f(x+h) = f(x) + f'(x)\frac{h}{1!} + f''(x)\frac{h^2}{2!} + f'''(x)\frac{h^3}{3!} + f''''(x)\frac{h^4}{4!} + \ldots$$

For $x = 0$,

$$g(0+h) = g(0) + g'(0)\frac{h}{1!} + g''(0)\frac{h^2}{2!} + g'''(0)\frac{h^3}{3!} + g''''(0)\frac{h^4}{4!} + \dots$$

$$\Rightarrow g(h) = 0 + \frac{h}{1!} + 0 - \frac{h^3}{3!} + 0 + \dots$$

$$\Rightarrow g(x) = \frac{x}{1!} - \frac{x^3}{3!} + \cdots \Rightarrow \sin(x) = \frac{x}{1!} - \frac{x^3}{3!} + \cdots = x - \frac{x^3}{6} + \dots$$

Now, for the original function $f(x) = x \cdot \operatorname{cosec}(x)$,

$$x \cdot \operatorname{cosec}(x) = x \cdot \frac{1}{x - \frac{x^3}{6} + \dots}$$

Performing algebraic long division, we get,

$$\begin{array}{r|l} x - \frac{x^3}{6} + \dots & \frac{1}{x} + \frac{x}{6} + \dots \\ \hline 1 + 0 & \\ \underline{1 - \frac{x^2}{6} + \dots} & \\ \frac{x^2}{6} + \dots & \\ \underline{\frac{x^2}{6} + \dots} & \\ 0 + \dots & \\ \vdots & \end{array}$$

Hence, we can write,

$$\Rightarrow x \cdot \operatorname{cosec}(x) = x \cdot \left(\frac{1}{x} + \frac{x}{6} + \dots\right) \qquad \text{[Quotient of the above algebraic long division.]}$$

$$= 1 + \frac{x^2}{6} + \dots$$

Rubric:

- 1 point for defining $g(x)$ and finding each ith order derivative $g^i(x)$.
- 1 point for correct values of $g(0)$ and each $g^i(0)$.
- 2 points for deriving the correct series of $g(x)$.
- 3 points for deriving the correct series of $f(x)$.

3. a) Compare and contrast the following interpolation methods: Direct method, Newton's Divided Difference method, and Lagrangian method. 4 (CO3) (PO2)

Solution:

Any 2 valid points of comparison will suffice. Some comparisons can be —

Criteria	Direct	Newton's Divided Difference	Lagrangian
Polynomial form	$f_n(x) = a_0 + a_1x + \cdots + a_nx^n$	$f_n(x) = b_0 + b_1(x-x_0) + \cdots + b_n(x-x_0)\dots(x-x_{n-1})$	$f_n(x) = \sum_{i=0}^{n}\left(\prod_{j=0,j\neq i}^{n}\frac{x-x_j}{x_i-x_j}\right)f(x_i)$
How coefficients are found	All coefficients a_i are found together by solving a system of $n+1$ equations.	Each b_i is found one at a time via divided differences.	No system is solved; the weights are written down directly from the data.
Adding a new data point	Whole system must be re-solved.	Previous b_i are reused; only the new term is added.	All weight functions must be recomputed.

Rubric:

- 2 points for each valid point of comparison.

b) Suppose there is an unknown function $f(x)$ and you are given 3 points $(x_0, f(x_0))$, $(x_1, f(x_1))$, and $(x_2, f(x_2))$ through which $f(x)$ passes. Using Newton's Divided Difference method, derive the representation of the 2nd order Newton's Divided Difference polynomial $f_2(x)$. 9 (CO4) (PO1)

Solution:

The 2nd order NDD interpolant is written in the form,

$$f_2(x) = b_0 + b_1(x - x_0) + b_2(x - x_0)(x - x_1)$$

Finding b_0**:** Putting $x = x_0$, the last two terms vanish, so

$$f_2(x_0) = b_0 = f(x_0) \quad \Rightarrow \quad b_0 = f(x_0)$$

Finding b_1**:** Putting $x = x_1$,

$$f(x_1) = b_0 + b_1(x_1 - x_0) = f(x_0) + b_1(x_1 - x_0)$$

$$\Rightarrow b_1 = \frac{f(x_1) - f(x_0)}{x_1 - x_0} = f[x_1, x_0]$$

Finding b_2**:** Putting $x = x_2$,

$$f(x_2) = f(x_0) + b_1(x_2 - x_0) + b_2(x_2 - x_0)(x_2 - x_1)$$

$$\Rightarrow b_2(x_2 - x_0)(x_2 - x_1) = \big(f(x_2) - f(x_0)\big) - \frac{f(x_1) - f(x_0)}{x_1 - x_0}(x_2 - x_0)$$

Writing $x_2 - x_0 = (x_2 - x_1) + (x_1 - x_0)$ on the right-hand side,

$$b_2(x_2 - x_0)(x_2 - x_1) = \big(f(x_2) - f(x_0)\big) - \frac{f(x_1) - f(x_0)}{x_1 - x_0}(x_2 - x_1) - \big(f(x_1) - f(x_0)\big)$$

$$= \big(f(x_2) - f(x_1)\big) - \frac{f(x_1) - f(x_0)}{x_1 - x_0}(x_2 - x_1)$$

Dividing both sides by $(x_2 - x_0)(x_2 - x_1)$,

$$b_2 = \frac{\dfrac{f(x_2) - f(x_1)}{x_2 - x_1} - \dfrac{f(x_1) - f(x_0)}{x_1 - x_0}}{x_2 - x_0} = \frac{f[x_2, x_1] - f[x_1, x_0]}{x_2 - x_0} = f[x_2, x_1, x_0]$$

Hence, the expanded form of the quadratic NDD interpolant is,

$$f_2(x) = f(x_0) + f[x_1, x_0](x - x_0) + f[x_2, x_1, x_0](x - x_0)(x - x_1)$$

Rubric:

- 1 point for writing $f_2(x)$ in the proper form.
- 1 point for the correct derivation of b_0.
- 2 points for the correct derivation of b_1.
- 4 points for the correct derivation of b_2.
- 1 point for the correct expanded form of $f_2(x)$.

c) In modern football analytics, the *Expected Goals* (G) of a shot is the probability that the shot results in a goal. Suppose, you have computed the average G of shots taken from various distances D (measured straight from the goal), from the last World Cup, as shown in Table 2. 6 (CO2) (PO1)

A striker takes a shot from $D = 16$ m. Estimate the G of this shot using the Direct method of interpolation and a second-order polynomial.

Table 2: Expected goals G as a function of shot distance D for Question 3.c

Distance, D (m)	6	11	14	20	30
Expected Goals, G	0.55	0.30	0.19	0.09	0.03

Solution:

For second-order (quadratic) interpolation, the polynomial is,

$$xG(D) = a_0 + a_1 D + a_2 D^2$$

Here the order $n = 2$, so we choose the nearest $n + 1 = 3$ points that bracket $D = 16$. These are $D_0 = 11$, $D_1 = 14$, $D_2 = 20$, with xG values 0.30, 0.19, and 0.09 respectively. This gives,

$$a_0 + 11a_1 + 121a_2 = 0.30$$
$$a_0 + 14a_1 + 196a_2 = 0.19$$
$$a_0 + 20a_1 + 400a_2 = 0.09$$

In matrix form,

$$\begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 1 & 11 & 121 \\ 1 & 14 & 196 \\ 1 & 20 & 400 \end{bmatrix}^{-1} \begin{bmatrix} 0.30 \\ 0.19 \\ 0.09 \end{bmatrix} = \begin{bmatrix} 1.045556 \\ -0.092222 \\ 2.22222 \times 10^{-3} \end{bmatrix}$$

Hence,

$$xG(D) = 1.045556 - 0.092222\, D + 2.22222 \times 10^{-3}\, D^2; \quad 11 \le D \le 20$$

At $D = 16$,

$$xG(16) = 1.045556 - 0.092222 \times 16 + 2.22222 \times 10^{-3} \times (16)^2 = 0.1389$$

So the shot from 16 m has an estimated $xG \approx 0.14$.

Rubric:

- 1 point for the correct polynomial representation.
- 1 point for choosing the correct nearest points.
- 2 points for the correct coefficient values.
- 2 points for the correct answer.

d) i. What is the Runge phenomenon? Explain why simply increasing the order of a polynomial interpolant does not necessarily improve the approximation, and how spline interpolation mitigates the issue. Use necessary illustration(s). 4 (CO3) (PO2)

Solution:

The **Runge phenomenon** is the unwanted oscillatory behaviour that appears near the ends of a high-order polynomial interpolant when it is fitted through equidistant data points. Increasing the order makes the oscillations worse, not better, so a higher-order polynomial can be a *worse* approximation than a lower-order one.

Consider Runge's function $f(x) = \dfrac{1}{1 + 25x^2}$. A 9th-order polynomial passing through 10 equidistant points swings wildly at the two ends, as shown below.

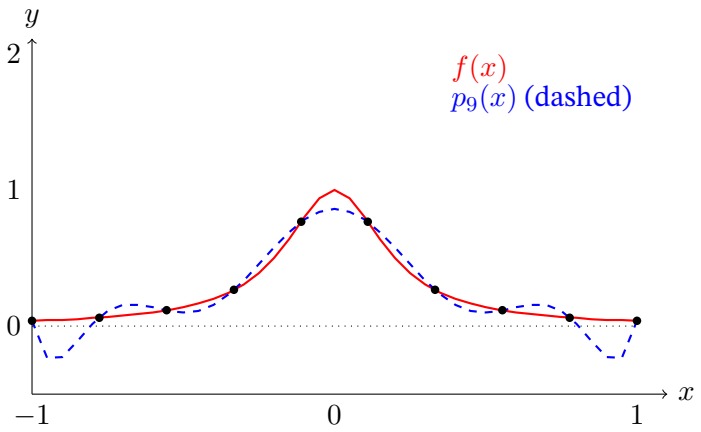


Figure 3: A 9th order polynomial interpolant through 10 equidistant points on Runge's function — note the oscillations near $x = \pm 1$ (bad).

A **spline** avoids this by fitting low-order (e.g. linear or cubic) pieces between consecutive points instead of one global high-order polynomial. Because each piece only spans a small interval, there is no room for large oscillations, so the spline stays faithful to the underlying function. This is why $S(x)$ tracks the data far better than $f_9(x)$.
Rubric:

- 1 point for mentioning the correct reason.
- 3 points for illustrating nice figure(s) explaining the Runge phenomenon.

ii. Suppose you want to perform Spline Interpolation on $n+1$ data points $(x_0, f(x_0))$, $(x_1, f(x_1)), \ldots, (x_n, f(x_n))$ using cubic splines of the form $f(x) = ax^3 + bx^2 + cx + d$. How many unknowns and how many equations would you have to deal with? Mention how you would obtain those equations and derive them. $1+4$ (CO4) (PO1)

Solution:

Each cubic spline has 4 unknowns. With $n+1$ points there are n consecutive intervals, hence n splines and $4n$ unknowns in total. To solve for them we need $4n$ equations.

The n splines forming the interpolant are,

$$f(x) = \begin{cases} a_1x^3 + b_1x^2 + c_1x + d_1; & x_0 \le x \le x_1 \\ a_2x^3 + b_2x^2 + c_2x + d_2; & x_1 \le x \le x_2 \\ \vdots & \vdots \\ a_nx^3 + b_nx^2 + c_nx + d_n; & x_{n-1} \le x \le x_n \end{cases}$$

(1) Interpolation conditions ($2n$ equations): Each spline passes through the two endpoints of its interval, so each interior point is shared by two splines. This gives $2n$ equations of the form

$$a_ix_{i-1}^3 + b_ix_{i-1}^2 + c_ix_{i-1} + d_i = f(x_{i-1})$$
$$a_ix_i^3 + b_ix_i^2 + c_ix_i + d_i = f(x_i)$$

(2) First-derivative continuity ($n-1$ equations): Consecutive splines must have equal slopes at each of the $n-1$ interior points:

$$3a_ix_i^2 + 2b_ix_i + c_i = 3a_{i+1}x_i^2 + 2b_{i+1}x_i + c_{i+1}$$

(3) Second-derivative continuity ($n-1$ equations): Consecutive splines must

also have equal curvature at each interior point:

$$6a_i x_i + 2b_i = 6a_{i+1}x_i + 2b_{i+1}$$

(4) Boundary conditions (2 equations): For a *natural* spline, the second derivative is set to zero at the two end points:

$$6a_1 x_0 + 2b_1 = 0 \qquad \text{and} \qquad 6a_n x_n + 2b_n = 0$$

In total, $2n + (n-1) + (n-1) + 2 = 4n$ equations for the $4n$ unknowns.

Rubric:

- 1 point for the correct number of unknowns and equations.
- 1 point for the $2n$ interpolation equations.
- 1 point for the $n-1$ first-derivative equations.
- 1 point for the $n-1$ second-derivative equations.
- 1 point for the 2 boundary equations.

MARKS SUMMARY TABLE

Question	Marks	Keywords
1.a	4	
1.b	6	
1.c)i	2	
1.c)ii	10	
2.a	4	
2.b)i	1	
2.b)ii	5	
2.c	15	
3.a	4	
3.b	9	
3.c	6	
3.d)i	4	
3.d)ii	5	
Total	75	